

123. Physical Medium Dependent (PMD) sublayer and medium, type 400GBASE-SR16

123.1 Overview

This clause specifies the 400GBASE-SR16 PMD together with the multimode fiber medium. The PMD sublayer provides a point-to-point 400 Gb/s Ethernet link over 16 pairs of multimode fiber, up to at least 100 m. When forming a complete Physical Layer, a PMD shall be connected to the appropriate PMA as shown in Table 123–1, to the medium through the MDI and optionally with the management functions that may be accessible through the management interface defined in Clause 45, or equivalent.

Table 123–1—Physical Layer clauses associated with the 400GBASE-SR16 PMD

Associated clause	400GBASE-SR16
117—RS	Required
117—400GMII ^a	Optional
118—400GMII Extender	Optional
119—PCS	Required
120—PMA	Required
120B—Chip-to-chip 400GAUI-16	Optional
120C—Chip-to-module 400GAUI-16	Optional
120D—Chip-to-chip 400GAUI-8	Optional
120E—Chip-to-module 400GAUI-8	Optional
78—Energy Efficient Ethernet	Optional

^a The 400GMII is an optional interface. However, if the 400GMII is not implemented, a conforming implementation behaves functionally as though the RS and 400GMII were present.

Figure 123–1 shows the relationship of the PMD and MDI (shown shaded) with other sublayers to the ISO/IEC Open System Interconnection (OSI) reference model. 400 Gb/s Ethernet is introduced in Clause 116 and the purpose of each PHY sublayer is summarized in 116.2.

400GBASE-SR16 PHYs with the optional Energy Efficient Ethernet (EEE) fast wake capability may enter the Low Power Idle (LPI) mode to conserve energy during periods of low link utilization (see Clause 78). The deep sleep mode of EEE is not supported.

Further relevant information may be found in Clause 1 (terminology and conventions, references, definitions and abbreviations) and Annex A (bibliography, referenced as [B1], [B2], etc.).

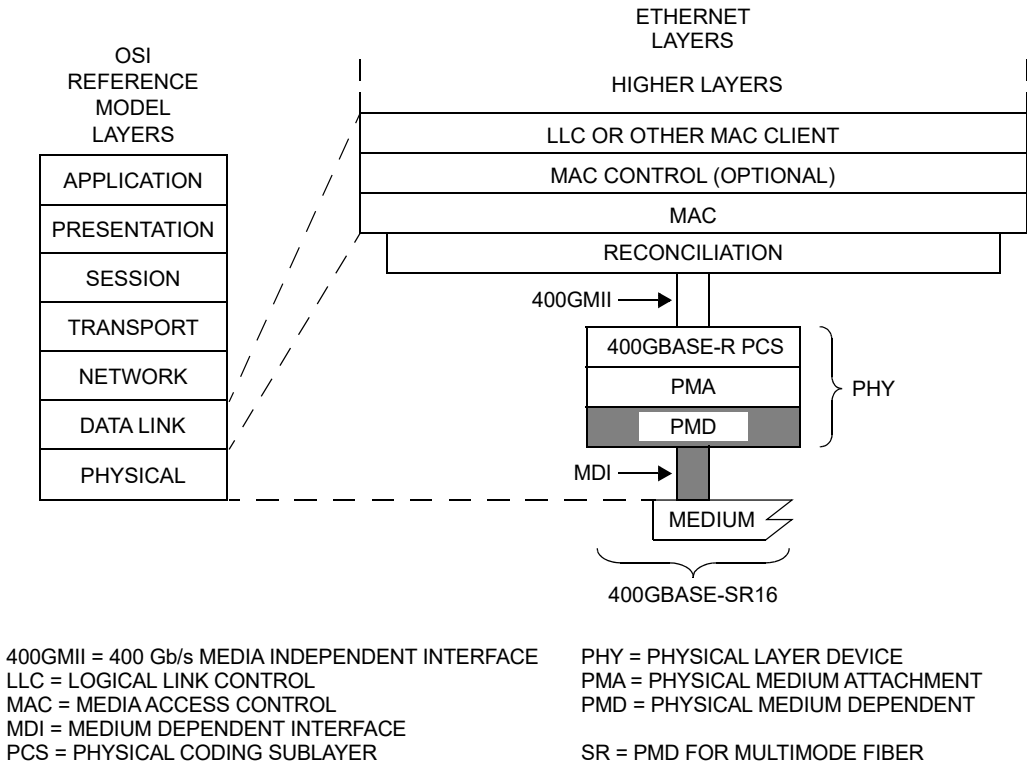


Figure 123-1—400GBASE-SR16 PMD relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and IEEE 802.3 Ethernet model

123.1.1 Bit error ratio

The bit error ratio (BER) shall be less than 2.4×10^{-4} provided that the error statistics are sufficiently random that this results in a frame loss ratio (see 1.4.344) of less than 1.7×10^{-12} for 64-octet frames with minimum interpacket gap when processed according to Clause 119. For a complete Physical Layer, the frame loss ratio may be degraded to 6.2×10^{-11} for 64-octet frames with minimum interpacket gap due to additional errors from the electrical interfaces.

If the error statistics are not sufficiently random to meet this requirement, then the BER shall be less than that required to give a frame loss ratio of less than 1.7×10^{-12} for 64-octet frames with minimum interpacket gap.

123.2 Physical Medium Dependent (PMD) service interface

This subclause specifies the services provided by the 400GBASE-SR16 PMD. The service interface for this PMD is described in an abstract manner and does not imply any particular implementation. The PMD service interface supports the exchange of encoded data between the PMA entity that resides just above the PMD, and the PMD entity. The PMD translates the encoded data to and from signals suitable for the specified medium.

The PMD service interface is an instance of the inter-sublayer service interface defined in 116.3. The PMD service interface primitives are summarized as follows:

PMD:IS_UNITDATA_*i*.request
PMD:IS_UNITDATA_*i*.indication
PMD:IS_SIGNAL.indication

The 400GBASE-SR16 PMD has 16 parallel bit streams, hence $i = 0$ to 15.

In the transmit direction, the PMA continuously sends 16 parallel bit streams to the PMD, one per lane, each at a nominal signaling rate of 26.5625 GBd. The PMD converts these streams of bits into appropriate signals on the MDI.

In the receive direction, the PMD continuously sends 16 parallel bit streams to the PMA corresponding to the signals received from the MDI, one per lane, each at a nominal signaling rate of 26.5625 GBd.

The SIGNAL_DETECT parameter defined in this clause maps to the SIGNAL_OK parameter in the PMD:IS_SIGNAL.indication(SIGNAL_OK) inter-sublayer service primitive defined in 116.3.

The SIGNAL_DETECT parameter can take on one of two values: OK or FAIL. When SIGNAL_DETECT = FAIL, the rx_symbol parameters are undefined.

NOTE—SIGNAL_DETECT = OK does not guarantee that the rx_symbol parameters are known to be good. It is possible for a poor quality link to provide sufficient light for a SIGNAL_DETECT = OK indication and still not meet the BER defined in 123.1.1.

123.3 Delay and Skew

123.3.1 Delay constraints

An upper bound to the delay through the PMA and PMD is required for predictable operation of the MAC Control PAUSE operation. The sum of the transmit and receive delays at one end of the link contributed by the 400GBASE-SR16 PMD including 2 m of fiber in one direction shall be no more than 8192 bit times (16 pause_quanta or 20.48 ns). A description of overall system delay constraints and the definitions for bit times and pause_quanta can be found in 116.4 and its references.

123.3.2 Skew constraints

The Skew (relative delay) between the lanes is kept within limits so that the information on the lanes can be reassembled by the PCS sublayer. Skew and Skew Variation are defined in 116.5 and specified at the points SP1 to SP6 shown in Figure 116-4 and Figure 116-5.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 43 ns and the Skew Variation at SP2 is limited to 400 ps.

The Skew at SP3 (the transmitter MDI) shall be less than 54 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 134 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 145 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation see 116.5. The measurements of Skew and Skew Variation are defined in 86.8.3.1.

123.4 PMD MDIO function mapping

The optional MDIO capability described in Clause 45 defines several variables that may provide control and status information for and about the PMD. If the MDIO interface is implemented, the mapping of MDIO control variables to PMD control variables shall be as shown in Table 123–2 and the mapping of MDIO status variables to PMD status variables shall be as shown in Table 123–3.

Table 123–2—MDIO/PMD control variable mapping

MDIO control variable	PMA/PMD register name	Register/bit number	PMD control variable
Reset	PMA/PMD control 1 register	1.0.15	PMD_reset
Global PMD transmit disable	PMD transmit disable register	1.9.0	PMD_global_transmit_disable
PMD transmit disable 15	PMD transmit disable extension register	1.27.0	PMD_transmit_disable_15
PMD transmit disable 14 to PMD transmit disable 0	PMD transmit disable register	1.9.15 to 1.9.1	PMD_transmit_disable_14 to PMD_transmit_disable_0

Table 123–3—MDIO/PMD status variable mapping

MDIO status variable	PMA/PMD register name	Register/bit number	PMD status variable
Fault	PMA/PMD status 1 register	1.1.7	PMD_fault
Transmit fault	PMA/PMD status 2 register	1.8.11	PMD_transmit_fault
Receive fault	PMA/PMD status 2 register	1.8.10	PMD_receive_fault
Global PMD receive signal detect	PMD receive signal detect register	1.10.0	PMD_global_signal_detect
PMD receive signal detect 15	PMD receive signal detect extension register	1.28.0	PMD_signal_detect_15
PMD receive signal detect 14 to PMD receive signal detect 0	PMD receive signal detect register	1.10.15 to 1.10.1	PMD_signal_detect_14 to PMD_signal_detect_0

123.5 PMD functional specifications

The 400GBASE-SR16 PMD performs the Transmit and Receive functions, which convey data between the PMD service interface and the MDI.

123.5.1 PMD block diagram

The PMD block diagram is shown in Figure 123–2. For purposes of system conformance, the PMD sublayer is standardized at the points described in this subclause. The optical transmit signal is defined at the output end of a multimode fiber patch cord (TP2), between 2 m and 5 m in length. Unless specified otherwise, all transmitter measurements and tests defined in 123.8 are made at TP2. The optical receive signal is defined at

the output of the fiber optic cabling (TP3) at the MDI (see 123.11.3). Unless specified otherwise, all receiver measurements and tests defined in 123.8 are made at TP3.

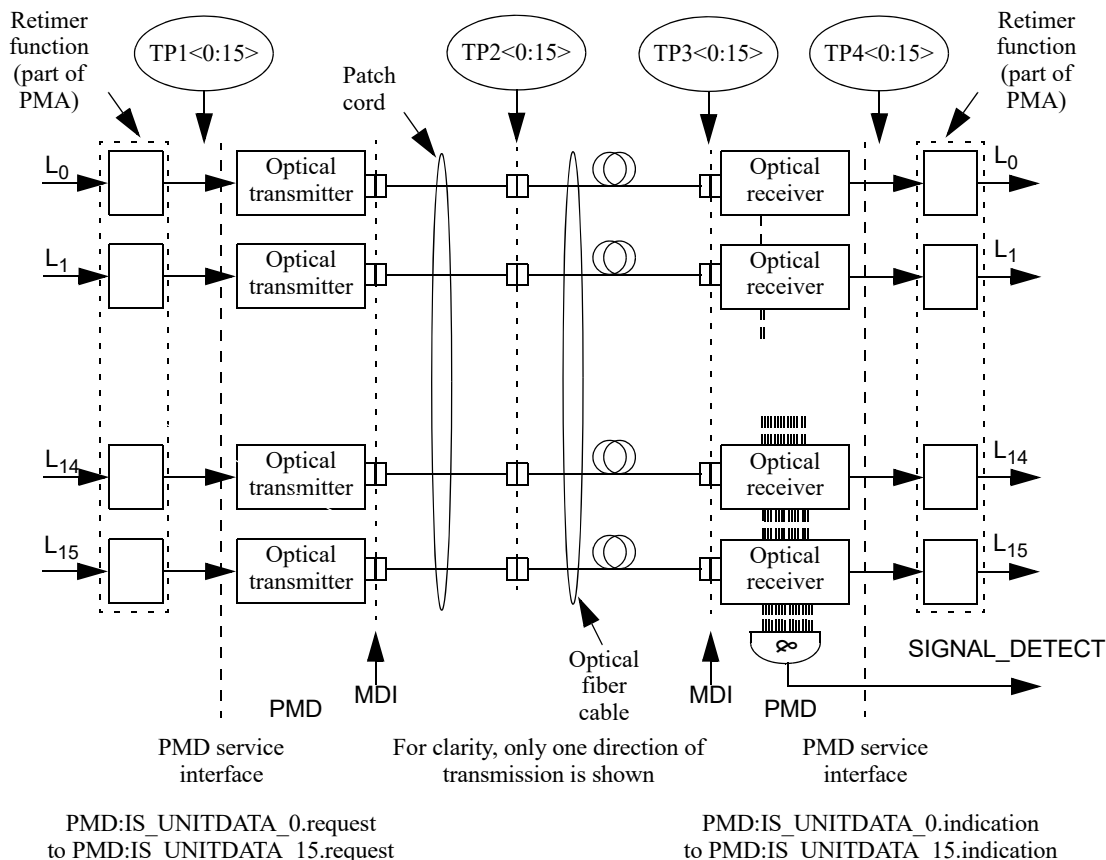


Figure 123–2—Block diagram for 400GBASE-SR16 transmit/receive paths

TP1 and TP4 are optional reference points that may be useful to implementers for testing components (these test points will not typically be accessible in an implemented system).

123.5.2 PMD transmit function

The PMD Transmit function shall convert the 16 bit streams requested by the PMD service interface messages `PMD:IS_UNITDATA_0.request` to `PMD:IS_UNITDATA_15.request` into 16 optical signals. The optical signals shall then be delivered to the MDI which contains 16 parallel light paths for transmit, according to the transmit optical specifications in this clause. The higher optical power level in each signal shall correspond to `tx_symbol = one`.

123.5.3 PMD receive function

The PMD Receive function shall convert the 16 optical signals received from the MDI into separate bit streams for delivery to the PMD service interface using the messages `PMD:IS_UNITDATA_0.indication` to `PMD:IS_UNITDATA_15.indication`, all according to the receive optical specifications in this clause. The higher optical power level in the signal shall correspond to `rx_symbol = one`.

123.5.4 PMD global signal detect function

The PMD global signal detect function shall report the state of SIGNAL_DETECT via the PMD service interface. The SIGNAL_DETECT parameter is signaled continuously, while the PMD:IS_SIGNAL.indication message is generated when a change in the value of SIGNAL_DETECT occurs. The SIGNAL_DETECT parameter defined in this clause maps to the SIGNAL_OK parameter in the inter-sublayer service interface primitives defined in 116.3.

SIGNAL_DETECT shall be an indicator of the presence of the optical signals on all 16 lanes. The value of the SIGNAL_DETECT parameter shall be generated according to the conditions defined in Table 123–4. The PMD receiver is not required to verify whether a compliant 400GBASE-SR16 signal is being received. This standard imposes no response time requirements on the generation of the SIGNAL_DETECT parameter.

Table 123–4—SIGNAL_DETECT value definition

Receive conditions	SIGNAL_DETECT value
For any lane; Average optical power at TP3 ≤ -30 dBm	FAIL
For all lanes; [(Optical power at TP3 $\geq$ average receive power, each lane (min) in Table 95–7) AND (compliant 400GBASE-SR16 signal input)]	OK
All other conditions	Unspecified

As a consequence of the requirements for the setting of the SIGNAL_DETECT parameter, implementations need to provide adequate margin between the input optical power level at which the SIGNAL_DETECT parameter is set to OK, and the inherent noise level of the PMD including the effects of crosstalk, power supply noise, etc.

Various implementations of the PMD global signal detect function are permitted by this standard, including implementations that generate the SIGNAL_DETECT parameter values in response to the amplitude of the modulation of the optical signals and implementations that respond to the average optical power of the modulated optical signals. When the MDIO is implemented, the SIGNAL_DETECT value shall be continuously set in response to the magnitude of the optical signals, according to the requirements of Table 123–4.

123.5.5 PMD lane-by-lane signal detect function

Various implementations of the Signal Detect function are permitted by this standard. When the MDIO is implemented, each PMD_signal_detect_*i*, where *i* represents the lane number in the range 0:15, shall be continuously set in response to the magnitude of the optical signal on its associated lane, according to the requirements of Table 123–4.

123.5.6 PMD reset function

If the MDIO interface is implemented, and if PMD_reset is asserted, the PMD shall be reset as defined in 45.2.1.1.1.

123.5.7 PMD global transmit disable function (optional)

The PMD global transmit disable function is optional and allows all of the optical transmitters to be disabled.

- a) When the `PMD_global_transmit_disable` variable is set to one, this function shall turn off all of the optical transmitters so that each transmitter meets the requirements of the average launch power of the OFF transmitter in Table 95–6.
- b) If a `PMD_fault` is detected, then the PMD may set the `PMD_global_transmit_disable` variable to one, turning off the optical transmitter in each lane.

123.5.8 PMD lane-by-lane transmit disable function (optional)

The PMD lane-by-lane transmit disable function is optional and allows the optical transmitter in each lane to be selectively disabled.

- a) When a `PMD_transmit_disablei` variable (where i represents the lane number in the range 0:15) is set to one, this function shall turn off the optical transmitter associated with that variable so that the transmitter meets the requirements of the average launch power of the OFF transmitter in Table 95–6.
- b) If a `PMD_fault` is detected, then the PMD may set each `PMD_transmit_disablei` variable to one, turning off the optical transmitter in each lane.

If the optional PMD lane-by-lane transmit disable function is not implemented in MDIO, an alternative method may be provided to independently disable each transmit lane.

123.5.9 PMD fault function (optional)

If the PMD has detected a local fault on any of the transmit or receive paths, the PMD shall set `PMD_fault` to one.

If the MDIO interface is implemented, `PMD_fault` shall be mapped to the fault bit as specified in 45.2.1.2.3.

123.5.10 PMD transmit fault function (optional)

If the PMD has detected a local fault on the transmitter, the PMD shall set `PMD_transmit_fault` to one.

If the MDIO interface is implemented, `PMD_transmit_fault` shall be mapped to the transmit fault bit as specified in 45.2.1.7.4.

123.5.11 PMD receive fault function (optional)

If the PMD has detected a local fault on the receiver, the PMD shall set the `PMD_receive_fault` variable to one.

If the MDIO interface is implemented, `PMD_receive_fault` shall be mapped to the receive fault bit as specified in 45.2.1.7.5.

123.6 Lane assignments

There are no lane assignments (within a group of transmit or receive lanes) for 400GBASE-SR16. While it is expected that a PMD will map electrical lane i to optical lane i and vice versa, there is no need to define the physical ordering of the lanes, as the PCS sublayer is capable of receiving the lanes in any arrangement. The positioning of transmit and receive lanes at the MDI is specified in 123.11.3.

123.7 PMD to MDI optical specifications for 400GBASE-SR16

The operating range for the 400GBASE-SR16 PMD is defined in Table 123–5. A 400GBASE-SR16 compliant PMD operates on 50/125 μm multimode fibers, type A1a.2 (OM3), type A1a.3 (OM4), or type A1a.4 (OM5), according to the specifications defined in Table 123–6. A PMD that exceeds the operating range requirement while meeting all other optical specifications is considered compliant (e.g., a 400GBASE-SR16 PMD operating at 120 m meets the operating range requirement of 0.5 m to 100 m). The operating range shown in Table 123–5 is the same as 100GBASE-SR4 (See Clause 95).

Table 123–5—400GBASE-SR16 operating range

PMD type	Required operating range ^a
400GBASE-SR16	0.5 m to 70 m for OM3
	0.5 m to 100 m for OM4
	0.5 m to 100 m for OM5

^a The PCS FEC correction function may not be bypassed for any operating distance.

123.7.1 400GBASE-SR16 transmitter optical specifications

The optical characteristics of each lane of a 400GBASE-SR16 transmitter shall be the same as those of a single lane of 100GBASE-SR4, as specified in Table 95–6 and 95.7.1, with the exception that the “signaling rate, each lane” is 26.5625 GBd $\pm$ 100 ppm, TDEC is as specified in 123.8.5, and the BER requirement is as specified in 123.1.1.

123.7.2 400GBASE-SR16 receive optical specifications

The optical characteristics of each lane of a 400GBASE-SR16 receiver shall be the same as those of a single lane of 100GBASE-SR4, as specified in Table 95–7 and 95.7.2, with the exception that the “signaling rate, each lane” is 26.5625 GBd $\pm$ 100 ppm, stressed receiver sensitivity is as specified in 123.8.8, and the BER requirement is as specified in 123.1.1.

123.7.3 400GBASE-SR16 illustrative link power budget

The illustrative power budget and penalties for each lane of a 400GBASE-SR16 link are the same as those of a single lane of 100GBASE-SR4, as specified in Table 95–8 and 95.7.3.

123.8 Definition of optical parameters and measurement methods

All transmitter optical measurements shall be made through a short patch cable, between 2 m and 5 m in length, unless otherwise specified. Optical parameters and measurement methods for 400GBASE-SR16 are the same as for 100GBASE-SR4.

123.8.1 Test patterns for optical parameters

The test patterns used in this clause shall be the same as those used for 100GBASE-SR4, as described in 95.8.1 and shown in Table 95–9, with the exception that pattern 5 defined in 82.2.11 is encoded by Clause 119 PCS for 400GBASE-SR16. Table 95–10 as modified by this subclause (123.8.1) shows the test patterns to be used in each measurement, unless otherwise specified, and also lists references to the subclauses in which each parameter is defined.

123.8.2 Center wavelength and spectral width

The center wavelength and RMS spectral width shall be within the range given in Table 95–6 if measured per IEC 61280-1-3. The transmitter is modulated using one of the test patterns specified in Table 95–10 as modified by 123.8.1.

123.8.3 Average optical power

The average optical power shall be within the limits given in Table 95–6 if measured using the methods given in IEC 61280-1-1. The average optical power is measured using one of the test patterns specified in Table 95–10 as modified by 123.8.1.

123.8.4 Optical Modulation Amplitude (OMA)

OMA shall be within the limits given in Table 95–6 if measured as defined in 52.9.5 for measurement with a square wave (8 ones, 8 zeros) test pattern or as defined in 68.6.2 (from the variable MeasuredOMA in 68.6.6.2) for measurement with a PRBS9 test pattern. See 123.8.1 for test pattern information.

123.8.5 Transmitter and dispersion eye closure (TDEC)

TDEC is a measure of the optical transmitter's vertical eye closure, as described in 95.8.5. TDEC shall be within the limits given in Table 95–6 if measured using the methods specified in 95.8.5.1 and 95.8.5.2, with the exceptions that the clock recovery unit (CRU) has a corner frequency of 4 MHz and in Equation (95–6), 3.8906R is replaced by 3.4917R, for consistency with the BER of 2.4×10^{-4} given in 123.1.1. Table 95–10 as modified by 123.8.1 specifies the test patterns to be used for measurement of TDEC.

123.8.6 Extinction ratio

The extinction ratio shall be within the limits given in Table 95–6 if measured using the methods specified in IEC 61280-2-2. The extinction ratio is measured using one of the test patterns specified for extinction ratio in Table 95–10 as modified by 123.8.1.

NOTE—Extinction ratio and OMA are defined with different test patterns (see Table 95–10).

123.8.7 Transmitter optical waveform (transmit eye)

The transmitter optical waveform of a port transmitting the test pattern specified in Table 95–10 as modified by 123.8.1 shall meet the transmitter eye mask coordinates and hit ratio specified in Table 95–6 if measured according to the methods specified in 95.8.7 with the exception that the clock recovery unit's high frequency corner bandwidth is 4 MHz.

123.8.8 Stressed receiver sensitivity

Stressed receiver sensitivity shall be within the limits given in Table 95–7 if measured using the methods defined in 95.8.8 with the following exceptions:

- The signaling rate of the test pattern generator is set to the rate defined in 123.7.2.
- The clock recovery unit (CRU) has a corner bandwidth of 4 MHz.
- Sinusoidal jitter is added as specified in Table 87–13 instead of Table 95–11.
- When using 95.8.5 to measure the SEC of the stressed receiver conformance test signal, the BER requirement is as specified in 123.1.1 and 3.4917R replaces 3.8906R in Equation (95–6).
- The hit ratio for the stressed receiver eye mask definition is changed from 5×10^{-5} to 2.4×10^{-4} .

123.9 Safety, installation, environment, and labeling

123.9.1 General safety

All equipment subject to this clause shall conform to IEC 60950-1.

123.9.2 Laser safety

400GBASE-SR16 optical transceivers shall conform to Hazard Level 1M laser requirements as defined in IEC 60825-1 and IEC 60825-2, under any condition of operation. This includes single fault conditions whether coupled into a fiber or out of an open bore.

Conformance to additional laser safety standards may be required for operation within specific geographic regions.

Laser safety standards and regulations require that the manufacturer of a laser product provide information about the product's laser, safety features, labeling, use, maintenance, and service. This documentation explicitly defines requirements and usage restrictions on the host system necessary to meet these safety certifications.²¹²

123.9.3 Installation

It is recommended that proper installation practices, as defined by applicable local codes and regulation, be followed in every instance in which such practices are applicable.

123.9.4 Environment

Normative specifications in this clause shall be met by a system integrating a 400GBASE-SR16 PMD over the life of the product while the product operates within the manufacturer's range of environmental, power, and other specifications.

It is recommended that manufacturers indicate, in the literature associated with the PHY, the operating environmental conditions to facilitate selection, installation, and maintenance.

It is recommended that manufacturers indicate, in the literature associated with the components of the optical link, the distance and operating environmental conditions over which the specifications of this clause will be met.

123.9.5 Electromagnetic emission

A system integrating a 400GBASE-SR16 PMD shall comply with applicable local and national codes for the limitation of electromagnetic interference.

123.9.6 Temperature, humidity, and handling

The optical link is expected to operate over a reasonable range of environmental conditions related to temperature, humidity, and physical handling (such as shock and vibration). Specific requirements and values for these parameters are considered to be beyond the scope of this standard.

²¹²A host system that fails to meet the manufacturer's requirements and/or usage restrictions may emit laser radiation in excess of the safety limits of one or more safety standards. In such a case, the host manufacturer is required to obtain its own laser safety certification.

123.9.7 PMD labeling requirements

It is recommended that each PHY (and supporting documentation) be labeled in a manner visible to the user, with at least the applicable safety warnings and the applicable port type designation (e.g., 400GBASE-SR16).

Labeling requirements for Hazard Level 1M lasers are given in the laser safety standards referenced in 123.9.2.

123.10 Fiber optic cabling model

The fiber optic cabling model is shown in Figure 123–3. The fiber type and length are the same as 100BASE-SR4 (see Clause 95).

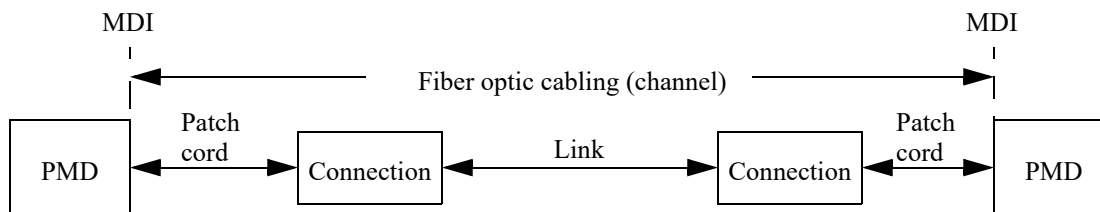


Figure 123–3—Fiber optic cabling model

The channel insertion loss is given in Table 123–6. A channel may contain additional connectors as long as the optical characteristics of the channel, such as attenuation, modal dispersion, reflections and losses of all connectors and splices meet the specifications. Insertion loss measurements of installed fiber cables are made in accordance with IEC 61280-4-1 one-cord method. As OM4 and OM5 optical fiber meet the requirements for OM3, a channel compliant to the “OM3” column may use OM4 or OM5 optical fiber, or a combination of OM3, OM4, and OM5. The fiber optic cabling model (channel) defined here is the same as a simplex fiber optic link segment. The term channel is used here for consistency with generic cabling standards.

Table 123–6—Fiber optic cabling (channel) characteristics for 400GBASE-SR16

Description	OM3	OM4	OM5	Unit
Operating distance (max)	70	100	100	m
Channel insertion loss ^a (max)	1.8	1.9	1.9	dB
Channel insertion loss (min)	0			dB

^a These channel insertion loss values include cable loss plus 1.5 dB allocated for connection and splice loss, over the wavelength range 840 nm to 860 nm.

123.11 Characteristics of the fiber optic cabling (channel)

The 400GBASE-SR16 fiber optic cabling shall meet the specifications defined in Table 123–6. The fiber optic cabling consists of one or more sections of fiber optic cable and any intermediate connections required to connect sections together. The requirements for the optical fiber, connection insertion loss, and maximum discrete reflectance are the same as 100GBASE-SR4 (see Clause 95).

123.11.1 Optical fiber cable

The fiber contained within the 400GBASE-SR16 fiber optic cabling shall comply with the specifications and parameters of Table 123–7. A variety of multimode cable types may satisfy these requirements, provided the resulting channel also meets the specifications of Table 123–6.

Table 123–7—Optical fiber and cable characteristics

Description	OM3 ^a	OM4 ^b	OM5 ^c	Unit
Nominal core diameter	50			μm
Nominal fiber specification wavelength	850			nm
Effective modal bandwidth (min) ^d	2000	4700		MHz.km
Cabled optical fiber attenuation (max)	3.5			dB/km
Zero dispersion wavelength (λ ₀)	1295 ≤ λ ₀ ≤ 1340		1297 ≤ λ ₀ ≤ 1328	nm
Chromatic dispersion slope (max) (S ₀)	0.105 for 1295 ≤ λ ₀ ≤ 1310 and 0.000375 × (1590 – λ ₀) for 1310 ≤ λ ₀ ≤ 1340		– 412/(840(1 – (λ ₀ /840) ⁴)) for 1297 ≤ λ ₀ ≤ 1328	ps/nm ² km

^a IEC 60793-2-10 type A1a.2.

^b IEC 60793-2-10 type A1a.3.

^c IEC 60793-2-10 type A1a.4.

^d When measured with the launch conditions specified in Table 95–6.

123.11.2 Optical fiber connection

An optical fiber connection, as shown in Figure 123–3, consists of a mated pair of optical connectors.

123.11.2.1 Connection insertion loss

The maximum link distance is based on an allocation of 1.5 dB total connection and splice loss. For example, this allocation supports three connections with an average insertion loss per connection of 0.5 dB. Connections with lower loss characteristics may be used provided the requirements of Table 123–6 are met. However, the loss of a single connection shall not exceed 0.75 dB.

123.11.2.2 Maximum discrete reflectance

The maximum discrete reflectance shall be less than –20 dB.

123.11.3 Medium Dependent Interface (MDI)

The 400GBASE-SR16 PMD is coupled to the fiber optic cabling at the MDI. The MDI is the interface between the PMD and the “fiber optic cabling” (as shown in Figure 123–3). Examples of an MDI include the following:

- a) Connectorized fiber pigtail
- b) PMD receptacle

123.11.3.1 Optical lane assignments

The 16 transmit and 16 receive optical lanes of 400GBASE-SR16 shall occupy the positions depicted in Figure 123-4 when looking into the MDI receptacle with the connector keyway feature on top. The interface contains 32 active lanes. The transmit optical lanes occupy the upper 16 positions. The receive optical lanes occupy the lower 16 positions.

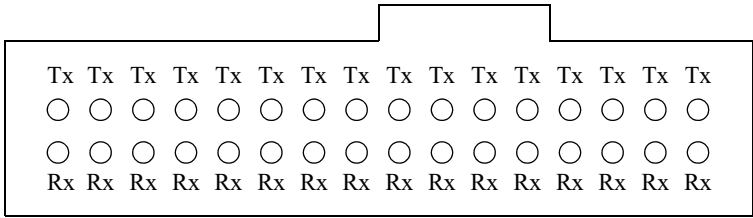


Figure 123-4—400GBASE-SR16 optical lane assignments

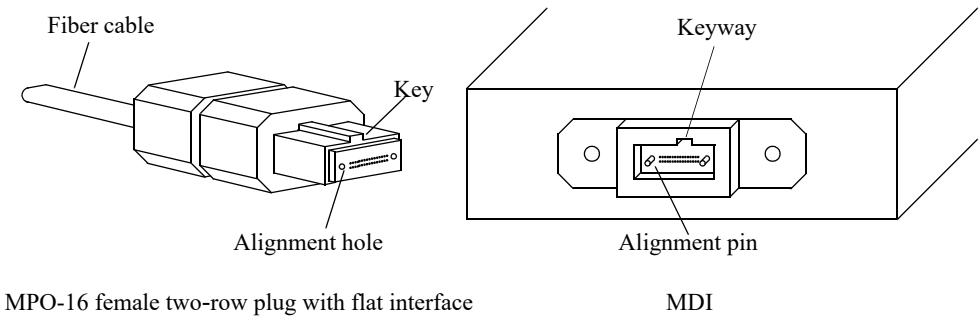


Figure 123-5—MPO-16 female two-row plug with flat interface, and an MDI

123.11.3.2 Medium Dependent Interface (MDI) requirements

The MDI adapter or receptacle shall meet the dimensional specifications for designation FOCIS 18 A-k-0 as defined in ANSI/TIA-604-18. The plug terminating the optical fiber cabling shall meet the dimensional specifications of designation FOCIS 18 P-2x16-1-0-2-2 as defined in ANSI/TIA-604-18. The MDI shall optically mate with the plug on the optical fiber cabling. Figure 123-5 shows an MPO-16 female two-row plug with flat interface, and an MDI. The MDI shall meet the interface performance specifications of IEC 61753-1 and IEC 61753-022-2 for performance Class Cm/2m.

NOTE—Transmitter compliance testing is performed at TP2 as defined in 123.5.1, not at the MDI.

123.12 Protocol implementation conformance statement (PICS) proforma for Clause 123, Physical Medium Dependent (PMD) sublayer and medium, type 400GBASE-SR16²¹³

123.12.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 123, Physical Medium Dependent (PMD) sublayer and medium, type 400GBASE-SR16, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

123.12.2 Identification

123.12.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

123.12.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3-2022, Clause 123, Physical Medium Dependent (PMD) sublayer and medium, type 400GBASE-SR16
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3-2022.)	

Date of Statement	
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²¹³ *Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

123.12.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
*INS	Installation/cable	123.10	Items marked with INS include installation practices and cable specifications not applicable to a PHY manufacturer	O	Yes [] No []
TP1	Reference point TP1 exposed and available for testing	123.5.1	This point may be made available for use by implementers to certify component conformance	O	Yes [] No []
TP4	Reference point TP4 exposed and available for testing	123.5.1	This point may be made available for use by implementers to certify component conformance	O	Yes [] No []
DC	Delay constraints	123.3.1	Device conforms to delay constraints	M	Yes []
SC	Skew constraints	123.3.2	Device conforms to Skew and Skew Variation constraints	M	Yes []
*MD	MDIO capability	123.4	Registers and interface supported	O	Yes [] No []

123.12.4 PICS proforma tables for Physical Medium Dependent (PMD) sublayer and medium, type 40GBASE-SR16

123.12.4.1 PMD functional specifications

Item	Feature	Subclause	Value/Comment	Status	Support
F1	Compatible with 40GBASE-R PCS and PMA	123.1		M	Yes []
F2	Integration with management functions	123.1		O	Yes [] No []
F3	Bit error ratio	123.1.1	Meets the BER specified in 123.1.1	M	Yes []
F4	Transmit function	123.5.2	Conveys bits from PMD service interface to MDI	M	Yes []
F5	Mapping between optical signal and logical signal for transmitter	123.5.2	Higher optical power is a one	M	Yes []
F6	Receive function	123.5.3	Conveys bits from MDI to PMD service interface	M	Yes []
F7	Conversion of 16 optical signals to 16 electrical signals	123.5.3	For delivery to the PMD service interface	M	Yes []
F8	Mapping between optical signal and logical signal for receiver	123.5.3	Higher optical power is a one	M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
F9	Global Signal Detect function	123.5.4	Report to the PMD service interface the message PMD:IS_SIGNAL.indication (SIGNAL_DETECT)	M	Yes []
F10	Global Signal Detect behavior	123.5.4	SIGNAL_DETECT is a global indicator of the presence of optical signals on all 16 lanes	M	Yes []
F11	Lane-by-lane Signal Detect function	123.5.5	Sets PMD_signal_detect_i values on a lane-by-lane basis per requirements of Table 123-4	MD:O	Yes [] No [] N/A []
F12	PMD reset function	123.5.6	Resets the PMD sublayer	MD:O	Yes [] No [] N/A []

123.12.4.2 Management functions

Item	Feature	Subclause	Value/Comment	Status	Support
M1	Management register set	123.4		MD:M	Yes [] N/A []
M2	Global transmit disable function	123.5.7	Disables all of the optical transmitters with the PMD_global_transmit_disable variable	MD:O	Yes [] No [] N/A []
M3	PMD lane-by-lane transmit disable function	123.5.8	Disables the optical transmitter on the lane associated with the PMD_transmit_disable_i variable	MD:O	Yes [] No [] N/A []
M4	PMD_fault function	123.5.9	Sets PMD_fault to one if any local fault is detected	MD:O	Yes [] No [] N/A []
M5	PMD_transmit_fault function	123.5.10	Sets PMD_transmit_fault to one if a local fault is detected on any transmit lane	MD:O	Yes [] No [] N/A []
M6	PMD_receive_fault function	123.5.11	Sets PMD_receive_fault to one if a local fault is detected on any receive lane	MD:O	Yes [] No [] N/A []

123.12.4.3 PMD to MDI optical specifications for 400GBASE-SR16

Item	Feature	Subclause	Value/Comment	Status	Support
SR1	Transmitter meets specifications in 95.7.1 with the exceptions in 123.7.1	123.7.1	Per definitions in 123.8	M	Yes []
SR2	Receiver meets specifications in 95.7.2 with the exceptions in 123.7.2	123.7.2	Per definitions in 123.8	M	Yes []

123.12.4.4 Optical measurement methods

Item	Feature	Subclause	Value/Comment	Status	Support
OM1	Measurement cable	123.8	2 m to 5 m in length	M	Yes []
OM2	Center wavelength and spectral width	123.8.2	Per IEC 61280-1-3 under modulated conditions	M	Yes []
OM3	Average optical power	123.8.3	Per IEC 61280-1-1	M	Yes []
OM4	OMA measurements	123.8.4	Each lane	M	Yes []
OM5	Transmitter and dispersion eye closure (TDEC)	123.8.5	Each lane	M	Yes []
OM6	Extinction ratio	123.8.6	Per IEC 61280-2-2	M	Yes []
OM7	Transmit eye	123.8.7	Each lane	M	Yes []
OM8	Stressed receiver sensitivity	123.8.8		M	Yes []

123.12.4.5 Environmental specifications

Item	Feature	Subclause	Value/Comment	Status	Support
ES1	General safety	123.9.1	Conforms to IEC 60950-1	M	Yes []
ES2	Laser safety—IEC Hazard Level 1M	123.9.2	Conforms to Hazard Level 1M laser requirements defined in IEC 60825-1 and IEC 60825-2	M	Yes []
ES3	Electromagnetic interference	123.9.5	Complies with applicable local and national codes for the limitation of electromagnetic interference	M	Yes []

123.12.4.6 Characteristics of the fiber optic cabling and MDI

Item	Feature	Subclause	Value/Comment	Status	Support
OC1	Fiber optic cabling	123.11	Meets requirements specified in Table 123–6	INS:M	Yes [] N/A []
OC2	Optical fiber characteristics	123.11.1	Per Table 123–7	INS:M	Yes [] N/A []
OC3	Maximum discrete reflectance	123.11.2.2	Less than –20 dB	INS:M	Yes [] N/A []
OC4	MDI layout	123.11.3.1	Optical lane assignments per Figure 123–4	M	Yes []
OC5	MDI dimensions	123.11.3.2	Meets FOCIS 18 A-k-0 as defined in ANSI/TIA-604-18	M	Yes []
OC6	Cabling connector dimensions	123.11.3.2	Meets FOCIS 18 P-2x16-1-0-2-2 as defined in ANSI/TIA-604-18	INS:M	Yes [] N/A []
OC7	MDI mating	123.11.3.2	MDI optically mates with plug on the cabling	M	Yes []

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Item	Feature	Subclause	Value/Comment	Status	Support
OC8	MDI requirements	123.11.3.2	Meets IEC 61753-1 and IEC 61753-022-2	INS:M	Yes [] N/A []